

Multilingual Machine Translation Hallucination Detection: Prompting Strategies and Refusal-Aware Evaluation

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Abstract

Multilingual Machine Translation (MT) has seen notable advances in recent years; however, hallucinations remain a critical challenge, particularly for Low-Resource Languages (LRLs). This paper evaluates hallucination detection strategies across 16 language directions in the HalOmi benchmark using a refusal-aware pipeline with two instruction-tuned models, Llama-3-8B and Qwen2.5-7B. We identify a “classification gap” in which models produce inaccurate hallucination labels under zero-shot prompting, but show improved performance with structured prompts and explicit reasoning. Our findings indicate that few-shot prompting, Chain-of-Thought (CoT), Chain-of-Translation (CoTR), and Multi-agent approaches can partially reduce this gap in High-Resource Languages (HRLs). However, advancements in LRLs are inconsistent, suggesting that prompting strategies alone cannot fully address the underlying multilingual limitations. Additionally, we find that model choice is a primary determinant of success: Qwen2.5-7B consistently outperforms Llama-3-8B in classification accuracy and response consistency, particularly for LRLs.

1 Introduction

Large Language Models (LLMs) have demonstrated exceptional fluency in multilingual generation and machine translation (MT) (Benkirane et al., 2024). However, they still tend to generate hallucinations, or fluent outputs that are not supported by the source text or semantically incorrect (Guerreiro et al., 2023a; Ji et al., 2023), particularly in non-English contexts (Kalai et al., 2025; Alansari and Luqman, 2025).

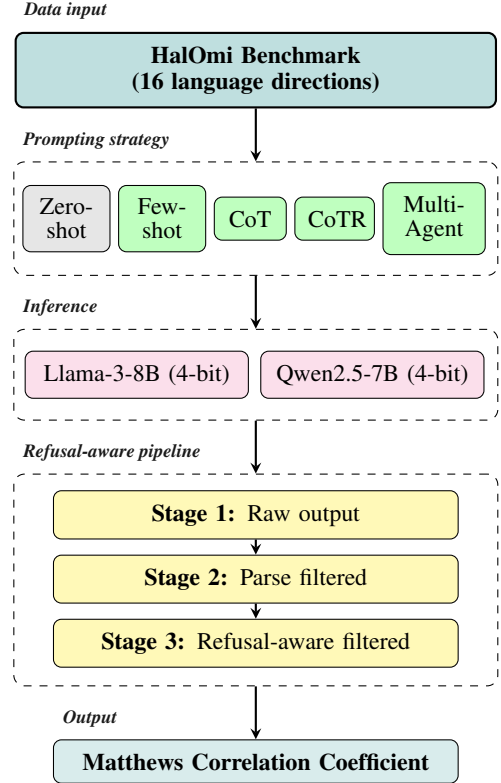


Figure 1: The hallucination detection pipeline integrates five prompting strategies across 16 language directions using Llama-3-8B and Qwen2.5-7B. A three-stage refusal-aware pipeline filters outputs to isolate raw performance from parse failures and safety-driven refusals. Performance is evaluated using the Matthews Correlation Coefficient (MCC) to account for the class imbalance in the *HalOmi* benchmark.

Recent studies have introduced LLM-as-a-Judge methods for automated hallucination detection (Li et al., 2024). However, these frameworks are primarily designed for high-resource languages (HRL), limiting their applicability to under-represented or low-resource languages (LRLs), where hallucinations are often more frequent. An LLM’s ability to accurately recall facts and generalise information is closely tied to the language being used; performance tends to decline sharply for languages that lack linguistic similarity with

English. Furthermore, traditional metrics for identifying hallucinations often rely on semantic similarity or embedding distance. These measures fail to capture cases where a model confidently generates completely unrelated information (Guerreiro et al., 2023a).

This paper investigates whether the detection of hallucinations in certain translation directions reflects genuine multilingual limitations or the effects of instruction tuning and prompt design. We specifically focus on an unexpected pattern where detection performance is weaker in some English-target directions than in lower-resource directions. To address this, we evaluate two small instruction-tuned models, Llama-3-8B (AI@Meta, 2024) and Qwen2.5-7B (Qwen Team, 2024), on the *HalOmi* benchmark (Dale et al., 2023) across 16 translation directions. We use a refusal-aware evaluation pipeline with Minos-v1 (Suphavadeeprasit et al., 2025) to separate explicit refusals from classification errors.

Additionally, we compare five prompting strategies: zero-shot, few-shot, Chain-of-Thought (CoT), Chain-of-Translation (CoTR), and Multi-agent review. Chain-of-Thought Self Consistency (CoT SC) is implemented as an additional experiment (Appendix D). This approach allows us to measure how different prompting strategies influence the cross-lingual reasoning failures in multilingual hallucination detection.

2 Related Work

Machine Translation Hallucinations and Detection Methods. Hallucinations in MT are outputs that are fluent but semantically unsupported by the source text (Ji et al., 2023; Huang et al., 2023). Prior work defines two main subtypes: *detached* hallucinations, where the output is entirely unrelated to the source, and *oscillatory* hallucinations, characterised by repetitive loops (Guerreiro et al., 2023b). Early approaches to hallucination detection relied on reference-based evaluation metrics or internal model signals, such as token-level uncertainty and entropy (Ji et al., 2023; Simhi et al., 2025). More recent methods without reference translations use cross-lingual embedding similarity with models such as LASER (Artetxe and Schwenk, 2019) and LaBSE (Feng et al., 2022), where low similarity between source and translation can indicate hallucination (Dale et al., 2022). Extending this analysis to more than

100 translation directions, Guerreiro et al. (2023b) found that hallucinations are more frequent in low-resource and non-English languages. Similarly, Islam et al. (2025) showed that smaller and multilingual models tend to hallucinate more often, especially in LRLs, and that stronger models often produce longer outputs in languages where they are more proficient, suggesting a trade-off between output length and hallucination counts rather than hallucination rates.

LLMs as Judges for MT Evaluation and Hallucination Detection. The LLM-as-a-Judge paradigm has substantially advanced automated evaluation. Zheng et al. (2023) showed that GPT-4 matched human preferences in over 80% of pairwise comparisons, while Liu et al. (2023) introduced G-Eval, which achieved strong correlation with human judgements in summarisation and outperformed prior automatic metrics. In MT, Kocmi and Federmann (2023) showed that prompted LLMs can achieve state-of-the-art quality assessment for HRLs without reference translations. Furthermore, Benkirane et al. (2024) applied LLM judges directly to multilingual MT hallucination detection, finding that while large open-weights models excelled in HRL settings, significantly larger proprietary systems achieved the highest detection accuracy for LRL directions. However, embedding-based methods remained competitive for non-Latin scripts, suggesting that the advantages of LLM judges are not uniform across scripts and resource levels.

Multilingual Challenges and Prompting Strategies. Multilingual hallucination detection remains challenging because non-Latin scripts can complicate tokenisation and source-target alignment for LLMs (Petrov et al., 2023), while cross-lingual generalisation remains uneven across languages (Hu et al., 2020). Therefore, prompting strategies have been explored to improve reliability in multilingual settings. In-context learning and few-shot prompting can provide clearer task instructions and output formats (Brown et al., 2020), while Chain-of-Thought (CoT) prompting encourages intermediate reasoning before a final decision (Wei et al., 2022). More recent approaches, such as translation-based decomposition and multi-agent review, aim to mitigate the classification gap in accuracy by breaking the model’s decision process into smaller reasoning steps (Su et al., 2025). Together, these methods motivate the

use of prompting strategies for multilingual hallucination detection.

3 Methods

3.1 Task Definition

We formulate hallucination detection as a binary classification problem. Given a source text S in language L_{src} and a machine-translated target text T in language L_{tgt} , the objective is to predict a label $Y \in \{0, 1\}$, where $Y = 0$ denotes the absence of hallucination and $Y = 1$ indicates the hallucination.

3.2 Data

We evaluate on *HalOmi* (Dale et al., 2023), a manually annotated benchmark for hallucination and omission detection in machine translation. The dataset covers 16 translation directions across high-resource languages (HRLs): Arabic (AR), Mandarin Chinese (ZH), German (DE), Russian (RU), and Spanish (ES), and low-resource languages (LRLs): Kashmiri (KS), Manipuri (MN), and Yoruba (YO), all paired with English (EN). *HalOmi* provides fine-grained annotations at multiple severity levels: *No Hallucination*, *Small*, *Partial*, and *Full*. Following Benkirane et al. (2024), we binarise these labels: *No Hallucination* is mapped to $Y = 0$, and all remaining severity levels to $Y = 1$. The resulting class distribution is imbalanced across translation directions; for example, in EN→RU, 141 out of 146 sentence pairs (96.6%) are labelled as *No Hallucination* (Appendix A). Therefore, we use the Matthews Correlation Coefficient (MCC) as our primary metric, as it handles imbalanced data well by balancing all correct and incorrect results across both labels, accounting for both minority and majority classes.

3.3 Models and Evaluation Pipeline

We experiment with two instruction-tuned models: Meta-Llama-3-8B-Instruct (AI@Meta, 2024) and Qwen2.5-7B-Instruct (Qwen Team, 2024), both run in 4-bit quantised form using the MLX-LM framework. Since *HalOmi* contains harmful content, models occasionally refuse to classify examples. To address this, we use the Minos-v1 classifier (Suphavadeeprasit et al., 2025) to distinguish two types of output failure (Figure 2): *parse failures*, from which no label can be extracted, and *refusals*, where the model avoids the task but may still yield a label. Because these failures fall dis-

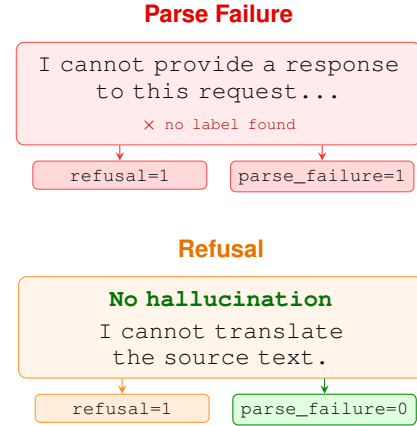


Figure 2: Output failure types identified using the Minos-v1 classifier. A parse failure (top) occurs when the model response contains no classification label, which is excluded from analysis. A refusal (bottom) occurs when the model provides a task label alongside a refusal message.

proportionately on hallucinated samples, we report results under three filtering stages (Figure 1).

3.4 Prompting Strategies

Our baseline is a zero-shot prompt adapted from Benkirane et al. (2024), which requests a binary label without intermediate reasoning (Figure 3). Beyond this baseline, we evaluate four strategies (Appendix C), each targeting a different hypothesised source of the classification gap:

Few-shot prompting prepends $k \in \{1, 2, 3\}$ labelled demonstrations from the same language pair before the test example, targeting output-format misalignment.

Chain-of-Thought (CoT) span-level detection adapts the framework of Su et al. (2025), prompting the model to reason step-by-step and identify hallucinated spans before assigning a label, targeting the lack of structured reasoning.

Chain-of-Translation (CoTR) follows Deshpande et al. (2024), projecting both source and target texts into English before classification to address cross-lingual processing difficulty.

Multi-Agent Review inspired by multi-agent debate frameworks (Du et al., 2023), gives each model the peer model’s baseline label.

4 Experiments

For each of the 16 translation directions, we performed inference on the full *HalOmi* test set using both models. The generation limit was set to 48 tokens for the baseline and few-shot prompts.

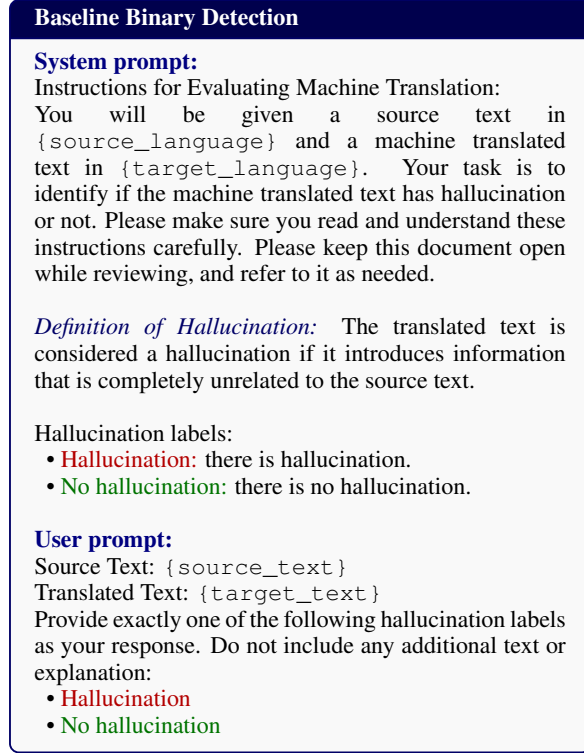
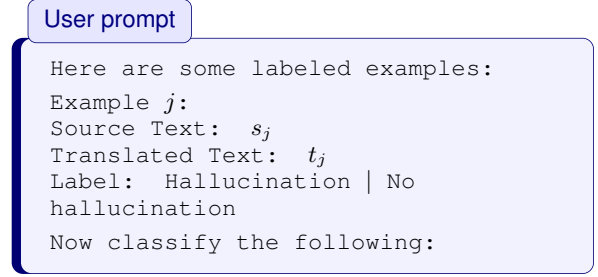


Figure 3: Baseline binary detection prompt (zero-shot), adapted from Benkirane et al. (2024). This prompt architecture establishes a zero-shot baseline by requiring the model to provide a binary classification without intermediate reasoning.

For prompts requiring intermediate reasoning, the limit was increased to 768 – 1024 tokens. Model outputs were processed for label extraction using regex, and refusal classification was performed post hoc with Minos-v1 (Suphavadeeprasit et al., 2025), with results reported across three filtering stages as outlined in Section 3.4.

4.1 Few-shot Prompting

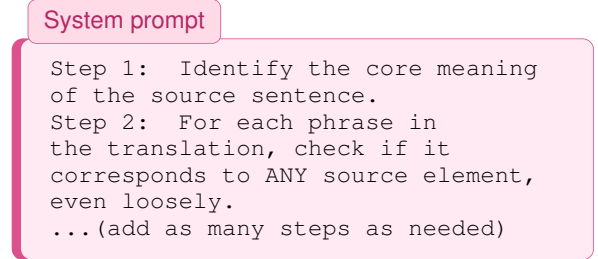
We investigate whether labeled in-context examples improve detection without additional reasoning. While zero-shot relies solely on task instructions, few-shot prepends $k \in \{1, 2, 3\}$ labeled demonstrations. For each test instance x_i , demonstrations are sampled uniformly at random from the same translation direction d , subject to leave-one-out: $x_i \notin \mathcal{D}_{\text{demo}}$. The random seed is fixed for reproducibility, and the natural class distribution is preserved. The system prompt s_{sys} is identical to the baseline (Figure 3, Appendix C, Figure 11). Demonstrations are prepended to the user prompt with numbered formatting:



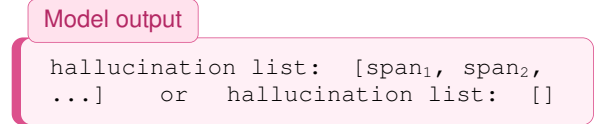
This yields a prompt $P_k = (s_{\text{sys}}, e_1, \dots, e_k, x_i)$, where $e_j = (s_j, t_j, y_j)$ denotes the j -th demonstration with source s_j , translation t_j , and label $y_j \in \{0, 1\}$. Results are evaluated against the $k = 0$ zero-shot condition.

4.2 Chain-of-Thought (CoT) Span-Level Detection

As simple in-context learning may not address specific semantics between source text and translation, we investigate whether structured reasoning allows models to better isolate hallucinated content. We adapt the span detection framework of Su et al. (2025) to machine translation, requiring the model to reason through sequential steps before producing a final label (Appendix C, Figure 12). The system prompt enforces the following reasoning structure:



A phrase is explicitly defined as *not* a hallucination if it translates or paraphrases any part of the source, even if the translation is incorrect (Figure 4). The model then outputs:



Binary labels are derived directly: a non-empty list implies hallucination ($\hat{y} = 1$), while an empty list indicates no hallucination ($\hat{y} = 0$).

4.3 Chain-of-Translation (CoTR) Prompting

While standard CoT provides a general reasoning framework, it may still be limited by the model’s ability to reason directly in low-resource languages. To overcome these inher-

ent reasoning bottlenecks, we evaluated Chain-of-Translation (CoTR) Prompting (Deshpande et al., 2024), which decomposes the task into a sequential translate-then-classify pipeline. This intervention elicits the model’s latent linguistic knowledge by projecting both the source and the machine-translated target into English before the final judgment phase. As detailed in Appendix C (Figure 13), the prompt is constructed based on the language pair:

1. **Source Projection:** If the source text S is in a non-English language L_{src} , the model first generates an English translation S_{en} .
2. **Target Projection:** If the machine-translated target T is in a non-English language L_{tgt} , the model generates an English translation T_{en} .
3. **Monolingual Classification:** The model compares the English representations (S_{en}, T_{en}) to determine if T contains information ungrounded in S .

This structured approach aims to bridge the classification gap by ensuring the discriminative phase occurs within a high-resource (English) context, mitigating the reasoning challenges in zero-shot binary classification.

4.4 Multi-Agent Review

Finally, we investigate whether label-level peer feedback can correct individual model errors. Prior work has shown that simple label aggregation accounts for most performance gains attributed to multi-agent debate (Choi et al., 2025); we therefore adopt a minimal variant that shares only binary labels, both to test whether peer signal alone can improve classification and to isolate the effect of label-level review from CoT reasoning, which is evaluated separately. We define two experimental conditions: In the first round, each model produces an independent label using the zero-shot baseline prompt (Figure 3). In the second round (R2), each model is provided with the peer model’s R1 label and prompted to reconsider its judgement. Unlike full debate frameworks that exchange reasoning chains, only the binary label is shared. To encourage genuine deliberation, the system prompt includes:

LLM	Subset	Base	Few-shot	CoT	CoTR	R2
Qwen2.5-7B	HRL	0.33	0.43	0.33	0.40	0.29
	LRL	0.13	0.20	0.12	0.12	0.14
LLaMA-3-8B	HRL	0.11	0.34	0.26	0.28	0.28
	LRL	0.08	0.13	0.09	0.12	0.11

Table 1: Mean MCC broken down by resource level (HRL: 10 directions; LRL: 6 directions) and prompting method.

LLM	Direction	Base	Few-shot	CoT	CoTR	R2
Qwen2.5-7B	EN $\rightarrow$ X	0.27	0.36	0.26	0.30	0.24
	X $\rightarrow$ EN	0.24	0.33	0.24	0.29	0.23
LLaMA-3-8B	EN $\rightarrow$ X	0.11	0.25	0.20	0.30	0.21
	X $\rightarrow$ EN	0.09	0.28	0.19	0.14	0.22

Table 2: Mean MCC broken down by direction type (EN $\rightarrow$ X: English as source; X $\rightarrow$ EN: English as target; 8 directions each) and prompting method.

System prompt

A peer model has already reviewed this example and provided their opinion. Consider it carefully, but make your own independent final judgement.

The peer model’s label is additionally included in the user prompt prior to the final answer instruction (Appendix C, Figure 14).

5 Results

We evaluate hallucination detection using MCC across 16 translation directions on the *HalOmi* dataset, comparing Qwen2.5-7B-Instruct and LLaMA-3-8B-Instruct. Tables 1-3 provide breakdowns by resource level, direction type, and model. Table 4 reports MCC excluding parse failures for all prompting strategies. Full results can be found in Appendix B, Tables 6-11.

Baseline Performance. Qwen2.5-7B outperforms LLaMA-3-8B across all language pairs except Manipuri (Appendix B.1, Table 7). On high-resource languages, Qwen achieves MCC ranging from 0.19 (English $\rightarrow$ Arabic) to 0.53 (English $\rightarrow$ Spanish), while LLaMA ranges from -0.11 (Arabic $\rightarrow$ English) to 0.36 (Spanish $\rightarrow$ English). LLaMA shows negative MCC on three HRLs (English $\rightarrow$ Arabic, Arabic $\rightarrow$ English, German $\rightarrow$ English), providing predictions worse than random. On low-resource languages, both models weaken:

LLM	Method	Mean MCC $\pm$ SD
QWEN2.5-7B	Baseline	0.25 $\pm$ 0.14
	Few-shot	0.34 $\pm$ 0.16
	CoT	0.25 $\pm$ 0.13
	CoTR	0.29 $\pm$ 0.17
	R2	0.24 $\pm$ 0.15
LLAMA-3-8B	Baseline	0.10 $\pm$ 0.13
	Few-shot	0.26 $\pm$ 0.14
	CoT	0.20 $\pm$ 0.10
	CoTR	0.22 $\pm$ 0.16
	R2	0.21 $\pm$ 0.12

Table 3: Model-level comparison across prompting strategies on *HalOmi*, reported as mean MCC $\pm$ standard deviation across 16 translation directions. Baseline uses MCC excluding parse failures. Few-shot reports the best MCC per direction over $k \in \{1, 2, 3\}$.

Qwen ranges from 0.05 (English $\rightarrow$ Manipuri) to 0.28 (English $\rightarrow$ Kashmiri), LLaMA from -0.07 (Yoruba $\rightarrow$ English) to 0.16 (Manipuri $\rightarrow$ English). LLaMA produces 50 refusals versus Qwen’s 3 (Suphavadeeprasit et al., 2025), disproportionately affecting low-resource directions.

Few-Shot Prompting. Few-shot prompting yields the highest mean MCC for both models (Table 3, Appendix B.2, Table 8): Qwen improves from 0.25 ± 0.14 to 0.34 ± 0.16 , LLaMA from 0.10 ± 0.13 to 0.26 ± 0.14 . For LLaMA on high-resource languages, 1-shot produces significant increase where baseline MCC is negative or near-zero (German $\rightarrow$ English: $-0.04 \rightarrow 0.45$; Chinese $\rightarrow$ English: $0.16 \rightarrow 0.36$), as shown in Table 8. These gains appear driven by correction of majority-class bias rather than improved detection (Brown et al., 2020). Qwen shows smaller improvements since baseline performance is already higher; peak values occur at varying shot counts (German $\rightarrow$ English 1-shot: $0.42 \rightarrow 0.55$; Arabic $\rightarrow$ English 2-shot: $0.29 \rightarrow 0.47$). On low-resource languages, few-shot benefits are minimal: Kashmiri $\rightarrow$ English decreases at 1-shot for Qwen ($0.14 \rightarrow -0.01$).

Chain-of-Thought Span Detection. CoT span-level detection provides modest improvements for LLaMA but minimal change for Qwen (Appendix B.3, Table 9). LLaMA benefits most on directions with low baselines (Arabic $\rightarrow$ English: $-0.11 \rightarrow 0.29$; English $\rightarrow$ Chinese: $0.03 \rightarrow 0.33$) (Wei et al., 2022). For Qwen, strong baseline directions can weaken (English $\rightarrow$ Spanish:

$0.53 \rightarrow 0.43$; German $\rightarrow$ English: $0.42 \rightarrow 0.31$). Low-resource languages show minimal benefit; Manipuri $\rightarrow$ English drops for Qwen ($0.11 \rightarrow 0.00$), suggesting that structured reasoning cannot overcome limited linguistic knowledge.

Chain-of-Thought (CoT) Span-Level Detection		
RU source	Плюнуть хочется в таких горе писателей статей!	
EN translation	I can't wait to see you write articles like this!	
Gold label	Hallucination	
Method	Prediction	Flagged span
Baseline	No hallucination $\times$	
CoT span	Hallucination $\checkmark$	I can't wait to see you write articles like this
Model reasoning: The source sentence is expressing a negative emotion towards the quality of writing, while the translation is expressing a positive emotion towards the act of writing.		

Figure 4: Example of CoT Span-level detection.

Chain-of-Translation. CoTR, which translates non-English inputs to English before classification (Bell et al., 2025), provides the largest improvement for LLaMA on high-resource languages (Appendix B.4, Table 10): English $\rightarrow$ Arabic improves from -0.03 to 0.56 , English $\rightarrow$ Russian from 0.24 to 0.40 . Qwen reaches peak performance on English $\rightarrow$ Spanish ($0.53 \rightarrow 0.61$) and Chinese $\rightarrow$ English ($0.29 \rightarrow 0.42$). However, CoTR decreases several low-resource directions: English $\rightarrow$ Manipuri drops for Qwen ($0.05 \rightarrow -0.02$), Kashmiri $\rightarrow$ English similarly ($0.14 \rightarrow 0.04$).

Multi-Agent Review. Multi-agent review (R2), where models exchange predictions before final classification, improves LLaMA’s mean MCC ($0.10 \rightarrow 0.21$) while Qwen remains stable ($0.25 \rightarrow 0.24$), as shown in Table 3 and Table 11. LLaMA shows high flip rates on low-resource directions (English $\rightarrow$ Kashmiri: 58%; Manipuri $\rightarrow$ English: 68%), demonstrating unstable baseline predictions. HRLs show minimal revision (English $\rightarrow$ Spanish: 4.6% for both models), suggesting that weaker models benefit from anchoring to stronger peers, while the reverse provides little or no increase.

6 Conclusion

This study evaluates hallucination detection across 16 translation directions using Llama-3-8B and Qwen2.5-7B with five prompting strategies. Several findings emerge from the results. Model selection plays an important role in the accu-

Direction	Qwen2.5-7B							LLaMA-3-8B						
	Base	1-shot	2-shot	3-shot	CoT	CoTR	R2	Base	1-shot	2-shot	3-shot	CoT	CoTR	R2
<i>HRL</i>														
EN → AR	0.19	0.35	0.30	0.31	0.18	0.29	0.17	-0.03	0.23	0.07	0.07	0.24	0.56	0.22
AR → EN	0.29	0.32	0.47	0.36	0.34	0.35	0.15	-0.11	0.21	0.17	0.00	0.29	0.08	0.26
EN → RU	0.27	0.38	0.28	0.20	0.30	0.42	0.25	0.24	0.22	0.25	0.19	0.19	0.40	0.27
RU → EN	0.19	0.23	0.26	0.26	0.24	0.23	0.21	0.09	0.32	0.20	0.30	0.19	0.06	0.14
EN → ZH	0.39	0.43	0.48	0.43	0.38	0.39	0.18	0.03	0.32	0.24	0.29	0.33	0.34	0.32
ZH → EN	0.29	0.36	0.37	0.34	0.35	0.42	0.31	0.16	0.36	0.24	0.14	0.23	0.27	0.27
EN → DE	0.28	0.29	0.18	0.14	0.32	0.36	0.29	0.13	0.39	0.29	0.24	0.28	0.30	0.15
DE → EN	0.42	0.55	0.53	0.50	0.31	0.46	0.44	-0.04	0.45	0.25	0.25	0.31	0.13	0.36
EN → ES	0.53	0.50	0.54	0.55	0.43	0.61	0.39	0.27	0.35	0.29	0.31	0.24	0.26	0.39
ES → EN	0.43	0.58	0.53	0.57	0.45	0.47	0.54	0.36	0.51	0.53	0.50	0.30	0.36	0.38
<i>LRL</i>														
EN → MNI	0.05	0.14	0.14	0.16	0.06	-0.02	0.17	0.06	0.05	0.10	0.07	0.06	0.07	0.02
MNI → EN	0.11	0.12	0.10	0.19	0.00	0.18	0.17	0.16	0.03	0.21	0.17	0.05	-0.04	0.11
EN → YO	0.11	0.20	0.19	0.25	0.17	0.17	0.16	0.06	0.05	0.12	0.17	0.05	0.10	0.06
YO → EN	0.07	0.06	0.03	0.02	0.09	0.15	-0.11	-0.07	0.01	-0.07	0.00	0.09	0.12	0.07
EN → KS	0.28	0.32	0.42	0.39	0.25	0.20	0.34	0.15	0.08	0.17	0.17	0.19	0.35	0.28
KS → EN	0.14	-0.01	-0.01	0.13	0.16	0.04	0.11	0.13	0.05	0.06	0.06	0.10	0.10	0.14

Table 4: MCC across all prompting strategies on *HalOmi* (Dale et al., 2023): zero-shot baseline, few-shot (1–3 shots), CoT span-level detection, Chain-of-Translation (CoTR), and multi-agent review (R2). Cell shading indicates **increase** or **decrease** in MCC relative to the per-direction baseline. Bold denotes the best value per direction per model.

racy of MT hallucination detection. Qwen2.5-7B achieves higher MCC than Llama-3-8B across most language pairs, with lower refusal rates (3 vs. 50) and more stable predictions (Table 3) and faster runtimes. Performance varies by resource level: both models show weaker results on LRLs than on HRLs (Table 1), and English-target directions generally yield higher MCC than non-English-target directions. Prompting strategies provide partial improvements. Few-shot prompting demonstrates the largest improvements, particularly for Llama-3-8B on directions with near-zero baseline scores. Chain-of-Thought (CoT) Span-level detection benefits Llama-3-8B on HRLs but shows inconsistent results for Qwen2.5-7B. Chain-of-Translation (CoTR) improves several HRL pairs by translating inputs into English before classification. Multi-agent review stabilises weaker model outputs but offers limited improvement for the stronger model. However, all strategies show reduced or inconsistent increases on LRLs, implying that prompting alone does not address underlying multilingual limitations (Benkirane et al., 2024). These results show that detection performance depends on model capacity,

language resource level, and translation direction. Structural interventions can mitigate the classification gap in HRLs, but their efficiency in LRLs remains limited.

7 Limitations

Our experiments used 4-bit quantised open-source models (7B-8B parameters) due to computational constraints. The results may differ for larger or proprietary systems (Benkirane et al., 2024). Hallucination detection accuracy was sensitive to prompt wording; minor phrasing changes produced negative MCC in some directions. The *HalOmi* dataset suffers from class imbalance, with > 90% non-hallucinated samples in several directions, limiting the reliability of comparisons. The dataset also contains noisy inputs (mixed-language text and markup fragments), which complicates distinguishing hallucinations from reasonable translations of distorted source text. Future work should evaluate whether larger models or reasoning-specialised systems reduce the classification gap. More robust prompt architectures may address the phrasing sensitivity observed in LRLs. Combining CoTR with multi-agent review could

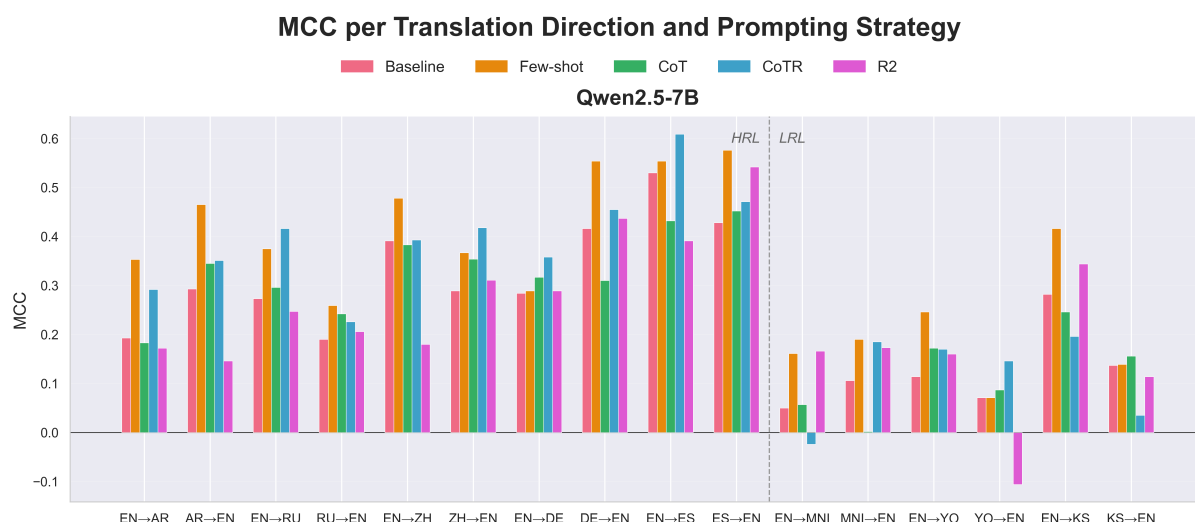


Figure 5: MCC for Qwen2.5-7B across all translation directions and prompting strategies.

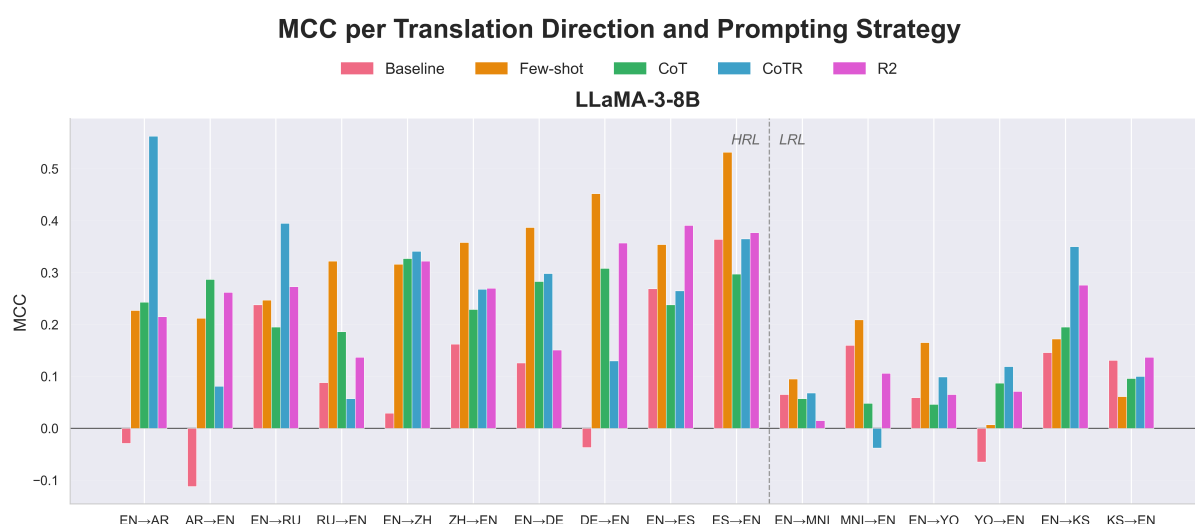


Figure 6: MCC for LLaMA-3-8B across all translation directions and prompting strategies.

improve reliability by leveraging English representations while reducing biases from safety refusals and imbalanced data.

8 Ethical Considerations

We evaluate hallucination detectors rather than deploy them, so direct misuse risks are limited. Reported MCC scores on Low-Resource Languages (LRLs) remain low, so treating these methods as reliable safeguards in Machine Translation (MT) pipelines is not warranted. *HalOmi* source texts may contain offensive or sensitive content; we use them unmodified for evaluation only.

Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic) in order to: Drafting con-

tent, grammar and spelling check. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

Licences

HalOmi (CC-BY-NC 4.0), Llama-3-8B-Instruct (Meta Llama 3 Community License), Qwen2.5-7B-Instruct and Minos-v1 (Apache 2.0) are used consistently with their non-commercial research terms.

References

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Appendix A Data

Appendix A.1 Dataset and Binary Labelling

The *HalOmi* benchmark (Dale et al., 2023) contains 2,562 sentence pairs across 16 English-centric translation directions covering 9 languages: Arabic (AR), Chinese (ZH), English (EN), German (DE), Kashmiri (KS), Manipuri (MN), Russian (RU), Spanish (ES), and Yoruba (YO). We adopt a binary labelling scheme in which all severity levels of hallucination (*Small*, *Partial*, *Full*) are merged into a *Hallucination* class, yielding 1,954 negative and 608 positive samples (23.7% hallucination rate overall). The per-direction label distribution is reported in Table 5.

Direction	<i>n</i>	No Hallucination	Hallucination
MN→EN	152	28.3%	71.7%
EN→MN	197	39.6%	60.4%
KS→EN	151	58.9%	41.1%
EN→KS	184	60.3%	39.7%
ES→EN	160	79.4%	20.6%
ZH→EN	159	79.9%	20.1%
EN→ZH	160	81.9%	18.1%
AR→EN	156	84.6%	15.4%
YO→EN	146	84.9%	15.1%
EN→YO	195	85.1%	14.9%
EN→ES	153	85.6%	14.4%
DE→EN	155	90.3%	9.7%
EN→DE	146	90.4%	9.6%
RU→EN	158	92.4%	7.6%
EN→AR	144	94.4%	5.6%
EN→RU	146	96.6%	3.4%
Total	2,562	76.3%	23.7%

Table 5: Binary label distribution across the 16 evaluated translation directions in the HALOMI benchmark. Proportions reflect a binary mapping where all hallucination severity levels are merged into a single positive class. The table is ordered by hallucination frequency, highlighting the significant label imbalance between low-resource pairs (e.g., MN, KS) and high-resource directions.

Appendix A.2 Hallucination Distribution

Figure 7 illustrates the distribution in hallucination prevalence across directions. LRL directions involving Manipuri and Kashmiri show rates between 40-72%, whereas HRL directions such as EN↔RU and EN↔DE remain below 10%.

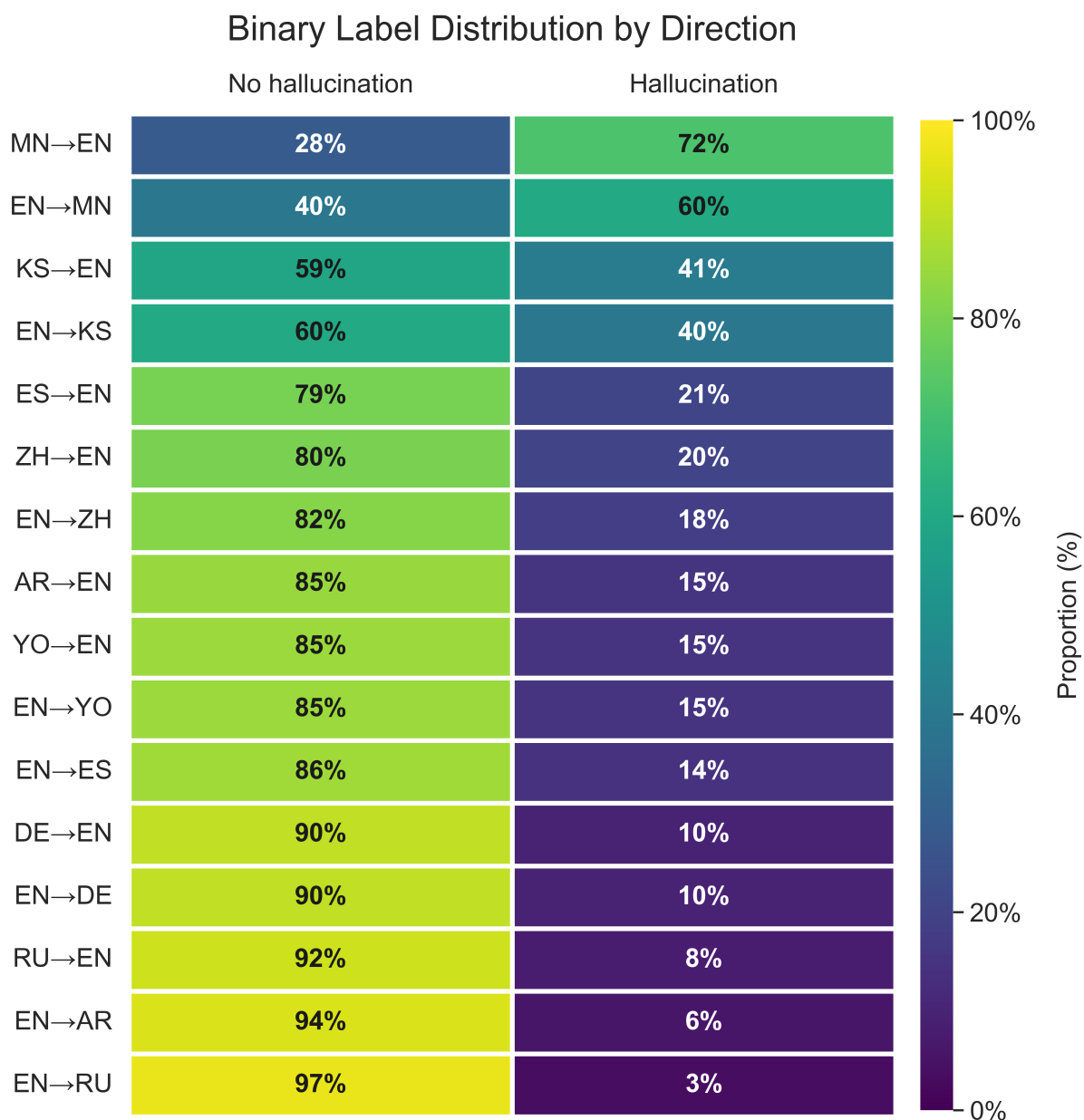


Figure 7: Heatmap of baseline hallucination prevalence across 16 language directions. The visualization illustrates the sharp contrast in translation reliability, with low-resource languages (LRLs) exhibiting hallucination rates up to 71.7%, while high-resource languages (HRLs) like German and Russian consistently remain below 10%.

Appendix B Results

This section (Tables 6-11) report MCC for all detection strategies across 16 language directions for both Qwen2.5-7B and LLaMA-3-8B: Zero-shot (Baseline), Few-shot prompting, Structured reasoning (CoT and CoTR), and Multi-agent review. Each method is evaluated under our refusal-aware pipeline Figure 1 (Suphavadeeprasit et al., 2025), which distinguishes raw from filtered scores by excluding parse failures and model refusals. MCC is used throughout to account for the label imbalance from *HalOmi* (Dale et al., 2023).

		Qwen2.5-7B		LLaMA-3-8B	
	Direction	Best Method	MCC	Best Method	MCC
<i>HRL</i>					
EN-AR	EN → AR	Few-shot	0.353	CoTR	0.563
	AR → EN	Few-shot	0.465	CoT	0.287
EN-RU	EN → RU	CoTR	0.416	CoTR	0.395
	RU → EN	Few-shot	0.259	Few-shot	0.322
EN-ZH	EN → ZH	Few-shot	0.478	CoTR	0.341
	ZH → EN	CoTR	0.418	Few-shot	0.358
EN-DE	EN → DE	CoTR	0.358	Few-shot	0.387
	DE → EN	Few-shot	0.554	Few-shot	0.452
EN-ES	EN → ES	CoTR	0.609	R2	0.391
	ES → EN	Few-shot	0.576	Few-shot	0.532
<i>LRL</i>					
EN-MNI	EN → MNI	R2	0.166	Few-shot	0.095
	MNI → EN	Few-shot	0.190	Few-shot	0.209
EN-YO	EN → YO	Few-shot	0.246	Few-shot	0.165
	YO → EN	CoTR	0.146	CoTR	0.119
EN-KS	EN → KS	Few-shot	0.416	CoTR	0.350
	KS → EN	CoT	0.156	R2	0.137

Table 6: Summary of the highest achieved MCC scores per direction. The table highlights which intervention (Few-shot, CoT, CoTR, or Multi-agent (R2)) best mitigated the classification gap for each model.

		Qwen2.5-7B			LLaMA-3-8B		
	Direction	MCC	MCC (Excl. Fail.)	MCC (Excl. Ref.)	MCC	MCC (Excl. Fail.)	MCC (Excl. Ref.)
<i>High-Resource Languages</i>							
English–Arabic	EN → AR	0.193	0.193	0.193	−0.029	−0.029	− 0.021
	AR → EN	0.293	0.293	0.293	− 0.112	−0.112	−0.112
English–Russian	EN → RU	0.273	0.273	0.273	0.172	0.238	0.000
	RU → EN	0.190	0.190	0.190	0.088	0.088	0.108
English–Chinese	EN → ZH	0.391	0.391	0.391	0.029	0.029	−0.038
	ZH → EN	0.296	0.289	0.289	0.162	0.162	0.162
English–German	EN → DE	0.284	0.284	0.284	0.126	0.126	0.229
	DE → EN	0.416	0.416	0.416	−0.037	−0.037	− 0.036
English–Spanish	EN → ES	0.530	0.530	0.530	0.269	0.269	0.126
	ES → EN	0.428	0.428	0.428	0.364	0.364	0.359
<i>Low-Resource Languages</i>							
English–Manipuri	EN → MNI	0.050	0.050	0.044	0.065	0.065	0.018
	MNI → EN	0.106	0.105	0.104	0.160	0.160	0.154
English–Yoruba	EN → YO	0.114	0.114	0.114	0.126	0.059	−0.025
	YO → EN	0.071	0.071	0.071	− 0.065	−0.065	−0.065
English–Kashmiri	EN → KS	0.282	0.282	0.282	0.146	0.146	0.170
	KS → EN	0.137	0.139	0.139	0.131	0.131	0.131
<i>Additional diagnostics</i>		Qwen2.5-7B			LLaMA-3-8B		
Parse failures		3			5		
Refusals		3			50		

Table 7: Baseline binary hallucination detection (MCC) for Qwen2.5-7B and LLaMA-3-8B (4-bit) across 16 translation directions on *HalOmi* (Dale et al., 2023), using the prompting strategy of Benkirane et al. (2024). Scores reported raw, excluding parse failures, and excluding refusals (Minos-v1; Suphavadeeprasit et al. (2025)). Bold indicates the best shot count per direction for each model.

Appendix B.2 Few-shot

		Qwen2.5-7B				LLaMA-3-8B			
	Direction	Base	1-shot	2-shot	3-shot	Base	1-shot	2-shot	3-shot
<i>High-Resource Languages</i>									
English–Arabic	EN → AR	0.193	0.353	0.300	0.314	−0.029	0.227	0.073	0.073
	AR → EN	0.293	0.317	0.465	0.360	−0.112	0.212	0.171	0.004
English–Russian	EN → RU	0.273	0.375	0.284	0.195	0.238	0.218	0.247	0.194
	RU → EN	0.237	0.232	0.259	0.259	0.079	0.322	0.198	0.299
English–Chinese	EN → ZH	0.391	0.433	0.478	0.432	0.029	0.316	0.236	0.289
	ZH → EN	0.289	0.360	0.367	0.342	0.162	0.358	0.239	0.139
English–German	EN → DE	0.284	0.289	0.184	0.143	0.126	0.387	0.288	0.241
	DE → EN	0.416	0.554	0.529	0.499	−0.037	0.452	0.249	0.249
English–Spanish	EN → ES	0.530	0.504	0.543	0.554	0.269	0.354	0.286	0.313
	ES → EN	0.403	0.576	0.528	0.574	0.348	0.507	0.532	0.504
<i>Low-Resource Languages</i>									
English–Manipuri	EN → MNI	0.050	0.140	0.143	0.161	0.065	0.050	0.095	0.067
	MNI → EN	0.105	0.115	0.102	0.190	0.160	0.032	0.209	0.165
English–Yoruba	EN → YO	0.114	0.197	0.187	0.246	0.059	0.046	0.115	0.165
	YO → EN	0.071	0.061	0.030	0.015	−0.059	0.007	−0.074	0.002
English–Kashmiri	EN → KS	0.282	0.324	0.416	0.392	0.146	0.081	0.172	0.170
	KS → EN	0.139	−0.013	−0.010	0.126	0.131	0.045	0.058	0.061

Table 8: Impact of in-context learning on detection accuracy across 1-shot to 3-shot conditions. The results demonstrate that while minimal labeled examples can effectively calibrate models for high-resource directions, the benefits for low-resource languages (e.g., Manipuri and Yoruba) are inconsistent.

		Qwen2.5-7B		LLaMA-3-8B	
	Direction	Baseline	CoT Span	Baseline	CoT Span
<i>High-Resource Languages</i>					
English–Arabic	EN → AR	0.1931	0.1829	−0.0288	0.2432
	AR → EN	0.2932	0.3445	−0.1116	0.2867
English–Russian	EN → RU	0.2731	0.2963	0.2379	0.1949
	RU → EN	0.1896	0.2416	0.0880	0.1863
English–Chinese	EN → ZH	0.3914	0.3826	0.0286	0.3272
	ZH → EN	0.2892	0.3536	0.1623	0.2290
English–German	EN → DE	0.2842	0.3170	0.1260	0.2825
	DE → EN	0.4161	0.3104	−0.0374	0.3077
English–Spanish	EN → ES	0.5296	0.4316	0.2685	0.2377
	ES → EN	0.4282	0.4515	0.3635	0.2970
<i>Low-Resource Languages</i>					
English–Manipuri	EN → MNI	0.0495	0.0569	0.0648	0.0567
	MNI → EN	0.1054	0.0020	0.1595	0.0480
English–Yoruba	EN → YO	0.1142	0.1722	0.0594	0.0457
	YO → EN	0.0707	0.0865	−0.0651	0.0872
English–Kashmiri	EN → KS	0.2822	0.2456	0.1457	0.1945
	KS → EN	0.1390	0.1558	0.1308	0.0964

Table 9: Comparative performance of the zero-shot baseline against structured Chain-of-Thought (CoT) reasoning. In this framework, models must explicitly extract source meaning and identify target discrepancies before issuing a label.

Appendix B.4 Chain-of-Translation (CoTR)

		Qwen2.5-7B		LLaMA-3-8B	
	Direction	Baseline	Translate	Baseline	Translate
<i>High-Resource Languages</i>					
English–Arabic	EN → AR	0.1931	0.2915	−0.0288	0.5632
	AR → EN	0.2932	0.3509	−0.1116	0.0814
English–Russian	EN → RU	0.2731	0.4162	0.2379	0.3954
	RU → EN	0.1896	0.2262	0.0880	0.0565
English–Chinese	EN → ZH	0.3914	0.3928	0.0286	0.3411
	ZH → EN	0.2892	0.4175	0.1623	0.2682
English–German	EN → DE	0.2842	0.3580	0.1260	0.2975
	DE → EN	0.4161	0.4553	−0.0374	0.1299
English–Spanish	EN → ES	0.5296	0.6094	0.2685	0.2647
	ES → EN	0.4282	0.4713	0.3635	0.3645
<i>Low-Resource Languages</i>					
English–Manipuri	EN → MNI	0.0495	−0.0241	0.0648	0.0675
	MNI → EN	0.1054	0.1846	0.1595	−0.0375
English–Yoruba	EN → YO	0.1142	0.1695	0.0594	0.0989
	YO → EN	0.0707	0.1460	−0.0651	0.1194
English–Kashmiri	EN → KS	0.2822	0.1958	0.1457	0.3504
	KS → EN	0.1390	0.0353	0.1308	0.1000

Table 10: Evaluation of the Chain-of-Translation (CoTR) strategy, which projects the evaluation task into an English-only context. By translating the non-English source or target into English before final judgment, the pipeline attempts to bypass cross-lingual reasoning challenges.

		Qwen2.5-7B			LLaMA-3-8B		
	Direction	Baseline	R2	Flip%	Baseline	R2	Flip%
<i>High-Resource Languages</i>							
English–Arabic	EN → AR	0.1931	0.1723	7.6	−0.0288	0.2150	10.4
	AR → EN	0.2932	0.1460	9.0	−0.1116	0.2621	25.0
English–Russian	EN → RU	0.2731	0.2471	8.2	0.2379	0.2731	11.6
	RU → EN	0.1896	0.2061	7.0	0.0880	0.1367	24.1
English–Chinese	EN → ZH	0.3914	0.1798	11.3	0.0286	0.3215	19.4
	ZH → EN	0.2892	0.3113	11.3	0.1623	0.2695	44.0
English–German	EN → DE	0.2842	0.2889	5.5	0.1260	0.1506	8.9
	DE → EN	0.4161	0.4371	4.5	−0.0374	0.3568	16.8
English–Spanish	EN → ES	0.5296	0.3907	4.6	0.2685	0.3907	4.6
	ES → EN	0.4282	0.5419	8.1	0.3635	0.3771	15.6
<i>Low-Resource Languages</i>							
English–Manipuri	EN → MNI	0.0495	0.1659	23.9	0.0648	0.0150	55.8
	MNI → EN	0.1054	0.1727	4.6	0.1595	0.1064	67.8
English–Yoruba	EN → YO	0.1142	0.1601	34.4	0.0594	0.0647	46.7
	YO → EN	0.0707	−0.1057	1.4	−0.0651	0.0707	48.6
English–Kashmiri	EN → KS	0.2822	0.3441	35.3	0.1457	0.2757	57.6
	KS → EN	0.1390	0.1140	2.7	0.1308	0.1373	49.0

Table 11: Multi-agent review results (R2) showing the impact of peer-label signaling. The **Flip%** indicates the frequency with which a model revised its initial judgment after observing a peer’s classification.

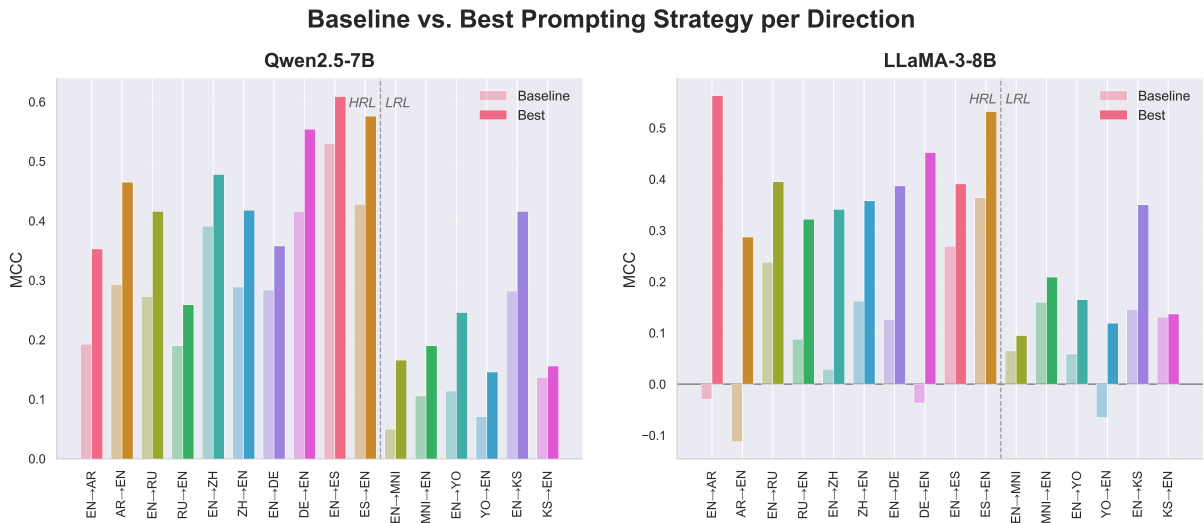


Figure 8: Baseline vs. Best Prompting Strategy (MCC)

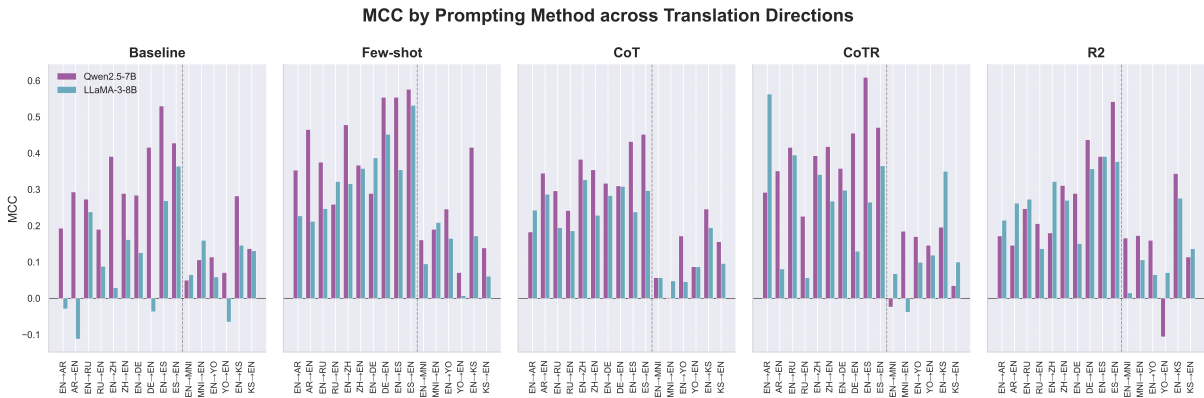


Figure 9: MCC by Prompting Method across Translational Directions

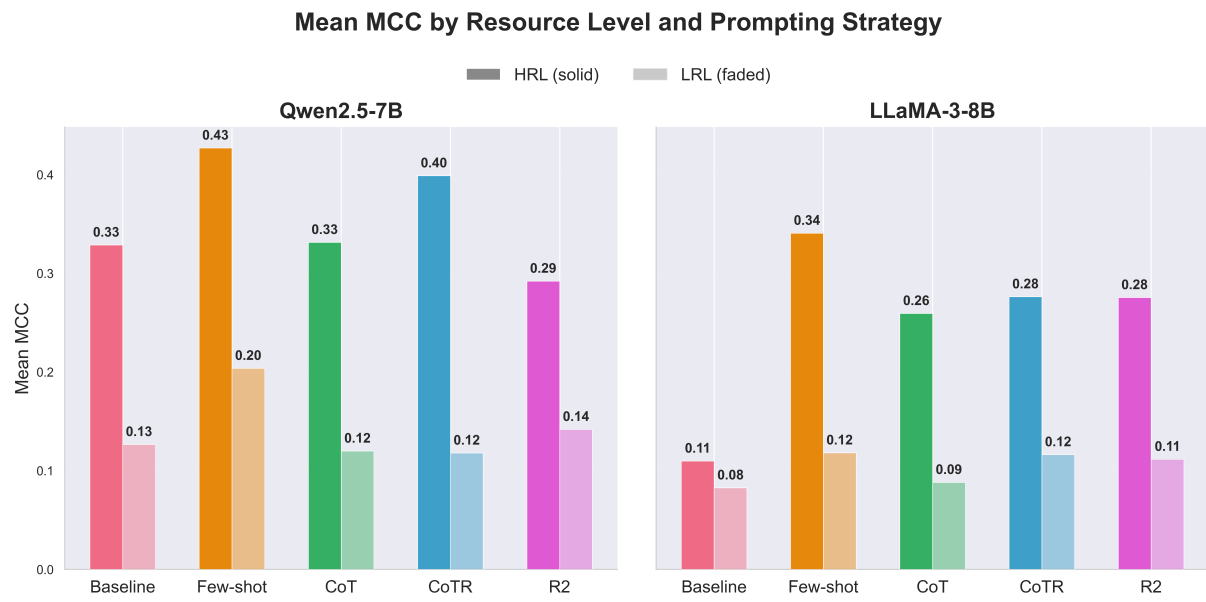


Figure 10: Mean MCC by Resource Level

Few-Shot Binary Detection

System prompt:
Instructions for Evaluating Machine Translation:
You will be given a source text in {source_language} and a machine translated text in {target_language}. Your task is to identify if the machine translated text has hallucination or not. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.
Definition of Hallucination: The translated text is considered a hallucination if it introduces information that is completely unrelated to the source text.
Hallucination labels:
• **Hallucination**: there is hallucination.
• **No hallucination**: there is no hallucination.

User prompt:
Here are some labeled examples:
Example 1:
Source Text: {demo_source_1}
Translated Text: {demo_target_1}
Label: {demo_label_1}
Example 2: ... (repeated for $k \in \{1, 2, 3\}$ demonstrations)
Now classify the following:
Source Text: {source_text}
Translated Text: {target_text}
Provide exactly one of the following hallucination labels as your response. Do not include any additional text or explanation:
• **Hallucination**
• **No hallucination**

Figure 11: Few-shot binary detection template. The user prompt is augmented with $k \in \{1, 2, 3\}$ in-context demonstrations from the same language direction to provide a calibration signal before the model issues a final binary classification.

Chain-of-Thought (CoT) Span-Level Detection

System prompt:
You will be given a source sentence in {source_language} and a machine-translated sentence in {target_language}. Your task is to determine whether the translation contains hallucination - content that has no basis in the source sentence. A phrase is NOT hallucination if it translates or paraphrases any part of the source, even if the translation is wrong. Hallucination means the phrase has NO corresponding source element at all. First, provide reasoning with the following format:
Step 1: Identify the core meaning of the source sentence.
Step 2: For each phrase in the translation, check if it corresponds to ANY source element, even loosely.
...(add as many steps as needed)
Then, compile the labeled hallucinated spans into a JSON dict, with a key hallucination list and its value is a list of hallucinated spans. If there are potential hallucinations, the output should be in the following JSON format:
hallucination list:
[hallucination span1, hallucination span2, ...].
Otherwise, leave the value as an empty list as follows: hallucination list: [].
End your response immediately after the hallucination list.

User prompt:
Source: {source_text}
Translation: {target_text}

Figure 12: Chain-of-Thought (CoT) span-level detection template. This structure requires the model to identify the source meaning and checking target phrases before outputting a JSON dictionary of specific hallucinated spans.

Chain-of-Translation (CoTR) Detection

System prompt:

Instructions for Evaluating Machine Translation:

Your task is to identify if a machine translated text contains hallucination or not.

Definition of Hallucination: The translated text is considered a hallucination if it introduces information that is completely unrelated to the source text.

Hallucination labels:

- **Hallucination**: there is hallucination.
- **No hallucination**: there is no hallucination.

User prompt:

Step 1: Translate the following {source_language} source text into English.

Step 2: Translate the following {target_language} machine translation into English.

Step 3: Compare the English translations and determine if the machine translation is a hallucination or not.

Source Text ({source_language}):
{source_text}

Machine Translation

({target_language}):
{target_text}

Follow the steps above. Show the English translation(s), then provide exactly one of the following labels on the final line:

Label: **Hallucination**

Label: **No hallucination**

Figure 13: Chain-of-Translation (CoTR) prompt architecture. The task is decomposed into English-language projections of the source and target texts, followed by a monolingual comparison phase to bypass the reasoning limitations inherent in low-resource language pairs.

Multi-Agent Review Prompt (R2)

System prompt:

Instructions for Evaluating Machine Translation:

You will be given a source text in {source_language} and a machine translated text in {target_language}.

Your task is to identify if the machine translated text contains a hallucination or not.

Definition of Hallucination: The translated text is a hallucination if it introduces information completely unrelated to the source text.

Hallucination labels:

- **Hallucination**: there is hallucination.
- **No hallucination**: there is no hallucination.

Peer Review: A peer model

({peer_name}) has already reviewed this example and provided their opinion.

Consider it carefully, but make your own independent final judgment.

User prompt:

Source Text: {source_text}

Translated Text: {target_text}

Peer ({peer_name}) label:
{peer_label_text}

Provide exactly one of the following hallucination labels as your response. Do not include any additional text or explanation:

- **Hallucination**
- **No hallucination**

Figure 14: Multi-agent review (R2) template used in the second inference round. The instruction incorporates a peer model's label (e.g., Qwen's label provided to Llama) to test if external signaling can elicit self-correction or improved classification confidence.

668 **Appendix D Additional Experiments**

669 We use Chain-of-Thought Self-Consistency (CoT-SC) (Wang et al., 2022) as an additional strategy for hallucination detection. The model generates multiple independent reasoning paths through higher-temperature sampling, each yielding a binary hallucination label. The final prediction is based on majority voting, which helps reduce output variance and improve reasoning stability. The prompt used is shown in Figure 15. Due to computational constraints, we evaluate CoT-SC only on Llama-3-8B for the Kashmiri-English directions. Results are reported in Table 12. MCC reaches 0.213 (EN→KS) and 0.184 (KS→EN), compared to baseline scores of 0.146 and 0.131 respectively. These findings should be treated as preliminary.

Direction	Llama-3-8B	
	Baseline	CoT-SC
EN → KS	0.146	0.213
KS → EN	0.131	0.184

Table 12: MCC scores for CoT-SC versus zero-shot baseline on the Kashmiri–English directions (Llama-3-8B). Results for additional language pairs and Qwen2.5-7B were not obtained due to computational constraints.

Chain-of-Thought Self-Consistency (CoT-SC)

System prompt:
Instructions for Evaluating Machine Translation Faithfulness:
You will be given:
• A source text in {source_language}
• A machine-translated text in {target_language}
Your task is to determine whether the translation contains hallucination.
Definition:
A hallucination occurs when the translation introduces content not supported by the source text, such as:
• Adding new facts or entities not present in the source
• Changing the meaning of the source
• Omitting critical information that alters the meaning
Important:
• Do NOT rely on word-by-word matching
• Differences in wording, grammar, or sentence structure are normal
• Paraphrasing and reordering are NOT hallucinations
• Function words or minor additions required for fluency are NOT hallucinations
Focus on whether the overall meaning of the translation is faithful to the source.
Decision Rule:
• If the translation preserves the meaning of the source → **No hallucination**
• If the translation introduces unsupported or incorrect meaning → **Hallucination**
User prompt:
Source Text: {source_text}
Translated Text: {target_text}
Please walk through the evaluation steps and provide your analysis.
End your response with exactly one final line in one of these formats:
Label: **Hallucination**
Label: **No hallucination**

Figure 15: Prompt used for Chain-of-Thought Self-Consistency (CoT-SC). Self-consistency is applied at inference time by sampling multiple reasoning paths and aggregating the final labels by majority vote.